

Koh Seungbeen

tact2021@gmail.com | Seoul, Republic of Korea

Research Summary

Protein Structural Biochemistry and Structure-Guided Therapeutic Discovery

Research Focus

My research experience has developed around protein structural biochemistry, with a focus on membrane receptors, receptor-ligand recognition, and structure-guided therapeutic discovery. Through undergraduate research, I have worked on GPCR expression and purification, receptor-ligand complex preparation, NMR-based peptide structural analysis, preliminary cryo-EM workflows, and biochemical validation assays. These experiences have trained me to think about biological questions through the relationship between molecular structure, biochemical function, and therapeutic relevance. They have also shaped my long-term interest in combining experimental structural biology with computational and AI-enabled approaches for protein design and drug discovery.

GPCR Structural Biochemistry: UR2R/hU2 Complex

My main research foundation was built in the Structural Biochemistry & Biophysics Laboratory at Yonsei University, where I led a first-author project on the human urotensin 2 receptor (UR2R)/urotensin 2 (hU2) complex. UR2R, also known as GPR14, is a class A GPCR associated with cardiovascular, metabolic, and disease-related signaling pathways. Because abnormal U2-UR2R signaling has been linked to pathological processes such as vasoconstriction, vascular remodeling, inflammation, metabolic dysregulation, and tissue fibrosis, UR2R is a biologically important and therapeutically relevant receptor system. However, structural information on the active UR2R/hU2 complex has remained limited, making biochemical preparation of the receptor complex an important step toward structure-guided drug discovery.

In this project, I focused on establishing an experimental workflow for producing ligand-bound UR2R and preparing the active receptor complex for preliminary cryo-EM characterization. I constructed human UR2R expression plasmids using the pFastBac/Bac-to-Bac system and performed baculovirus amplification, Sf9 insect cell expression, membrane preparation, detergent solubilization, TALON affinity purification, reverse Ni-NTA chromatography, HRV 3C cleavage, and ÄKTA-based size-exclusion chromatography. I also assembled the UR2R¹¹⁻³³⁶-hU2-mGqi-scFv16 signaling complex for cryo-EM sample preparation and contributed to preliminary cryo-EM characterization, including grid preparation, data collection/processing, and 2D class analysis as part of a collaborative workflow. Although the cryo-EM analysis remained preliminary, this project gave me direct experience with the practical challenges of membrane protein structural biology, including receptor expression, purification yield, sample homogeneity, complex stability, and particle behavior. I also organized experimental data, prepared figures, and contributed to manuscript writing, resulting in a first-author publication in *Protein Expression and Purification*.

Somatostatin Receptor Selectivity and NMR-Based Peptide Analysis

In parallel, I led a project on somatostatin analog structural analysis and receptor selectivity, which was presented as a first-author poster at ICMRBS 2024. While the UR2R project trained me in membrane receptor purification and complex preparation, this project expanded my perspective toward ligand structure and receptor-recognition mechanisms. Somatostatin is a cyclic peptide hormone that regulates neurotransmission and hormone secretion through GPCR-family somatostatin receptors. Since SSTR signaling is involved in endocrine regulation, cancer, and neurological disease, understanding how somatostatin and its analogs selectively interact with receptor subtypes is important for therapeutic design.

For this project, I participated in NMR sample measurements and analyzed HSQC, NOESY, and TOCSY spectra of designed somatostatin analogs using POKY. I performed peptide structure calculation, compared the conformations of designed analogs with SST-14, generated PyMOL-based structural figures, and interpreted SSTR2 docking models. By integrating NMR-derived peptide structures, receptor-ligand interaction analysis, sequence alignment, and HADDOCK-based docking, I investigated structural features associated with SSTR selectivity. A central insight from this work was that the conserved Trp-Lys motif of SST-14 appears to play a key role in receptor binding, while more variable receptor-interacting regions may contribute to isoform-specific ligand recognition. Docking analysis further

suggested that the designed analog SBMB-1 may bind SSTR2 in a manner similar to SST-14, supporting the potential of structure-guided analog design for developing SSTR-targeted agonists or antagonists. This project strengthened my interest in using structural information to understand ligand selectivity and guide therapeutic molecule design.

Additional GPCR and Drug Discovery Experience

I also contributed to a draft-stage project on human somatostatin receptor 5 (SSTR5) structural characterization. SSTR5 is another GPCR involved in endocrine regulation, neurological disease, hormone disorders, and cancer, yet structural information on SSTR5 remains relatively limited compared with other SSTR subtypes. My role in this project was more limited than in the UR2R and somatostatin analog projects, but I contributed to SSTR5 purification workflows and supported the organization and interpretation of purification and complex-formation data for SSTR5-G α i-scFv16 complexes.

Beyond GPCR structural biochemistry, I gained additional exposure to drug discovery and epigenetic regulation through short-term research experiences. At Epinogen, a Yonsei University-affiliated AI drug discovery company, I assisted with ASH1 protein purification and evaluated the effect of ASH1-targeting compound treatment by western blot analysis. Although my role was limited due to the short internship period, this experience introduced me to biochemical validation workflows in an AI-assisted epigenetic drug discovery setting and helped me connect protein purification and assay-based validation with therapeutic development.

Chromatin Biology and Epigenomic Profiling

I further broadened my experimental background through a research training program in the Epigenetics and Molecular Pharmacology Laboratory at Seoul National University. In one project, I worked on H3K27M-mediated PRC2 inhibition in diffuse intrinsic pontine glioma. I reconstituted WT, H3K27A, and H3K27M histone octamers, purified H3K27A/H3K27M octamers using ÄKTA-based size-exclusion chromatography, validated fractions by SDS-PAGE and Bradford assay, and assisted with PRC2-dinucleosome interaction and HMT assay workflows. In a second project, I performed CUT&Tag experiments using male and female mouse cortex and hippocampus samples to profile H3K4me3- and H3K27me3-associated chromatin states, followed by qPCR-based library quality control using HoxA9, Oct4, and indexed primer sets. These projects exposed me to chromatin biochemistry and epigenomic profiling, broadening my view of how biochemical and molecular assays can be used to study disease-related gene regulation.

Future Direction

Taken together, my research experiences have given me a strong foundation in experimental protein structural biochemistry, especially in GPCR and receptor-ligand systems, while also exposing me to peptide structural analysis, preliminary cryo-EM workflows, epigenetic drug discovery, and chromatin biology. The common theme across these projects is my interest in connecting molecular structure to biological function and therapeutic application. Moving forward, I hope to develop as a researcher who can combine rigorous experimental methods with computational and AI-enabled approaches to design, characterize, and optimize proteins or protein-targeting molecules with therapeutic potential.